Phoenix Autocallable Notes on the CAC40 under a Calibrated Heston Model: Fair Coupons, Scenarios and Greeks

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Abstract

We price single-underlying Phoenix autocallable notes, whose coupon is paid at each observation date where the index is above a coupon barrier, with or without memory of the missed coupons, on the CAC40 of 12 February 2025. The underlying follows a Heston model calibrated to the implied volatility surface (RMSE 0.42 volatility point), with a drift read from a forward curve bootstrapped from put–call parity. The note is described by a single parametrisation covering the observation frequency, coupon barrier and memory, autocall barrier and step-down, non-call period, European, American continuous or American daily capital protection, airbag and capital guarantee. The notes are priced by Monte Carlo with antithetic variates and a control variate; the American protection is handled with the Brownian bridge crossing probability as a conditional expectation. The pricer recovers two closed-form limit cases and an exact identity with the snowball Athena. We report the fair coupon (price at par) of 19 variants together with their scenario statistics: the base Phoenix (4 years, quarterly observations, memory, coupon barrier at 70%, autocall at 100%, European protection at 60%) pays 6.04% a year, is called after one quarter with probability 55.4% and loses capital with probability 5.2%. The memory is worth 0.37 point of coupon and an American protection costs 1.14 points. The note is almost Delta-neutral at the money, short volatility (-0.33 per vol point on a nominal of 100) and short dividends (-0.32 per point of carry).

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1 Introduction

In an AutoCallable note, the investor receives coupons well above the risk-free rate, and the note is redeemed early ("called") as soon as the index is above a reference level on an observation date. The coupon is financed by selling a down-and-in put. If the note is never called and the index ends below a protection barrier, the investor bears the loss of the index.

This report studies the **Phoenix**, whose coupon is paid at each observation date where the index is above a coupon barrier, lower than the autocall barrier, usually with a *memory* of the missed coupons. The other main family, the Athena, whose coupon is paid only when the note is called, is treated in a companion report [1]. On top of this structure, issuers combine many features: observation frequency, step-down of the autocall barrier, non-call period, European or American protection, airbag, capital guarantee. We will have a look on how to price these different versions in a single framework, on real market data, and measure how each feature moves the fair coupon, the probabilities of the scenarios and the Greeks.

The model is the Heston model calibrated in [2] to the CAC40 implied volatility surface, and the Monte Carlo engine is the one used for barrier options in [3]. The scripts and notebooks are open source; Section 7 describes the code.

2 Products

2.1 Common structure

A note of nominal N on an index S has an initial fixing S_{ref} and observation dates $0 < t_1 < \cdots < t_n = T$. The performance at t_k is $P_k = S_{t_k}/S_{\text{ref}}$. All barriers are fractions of S_{ref}: autocall barriers B_k^{call}, coupon barriers B_k^{cpn} and protection barrier B^{prot}. The dates t_k with $t_k \geq t_{\text{nc}}$ are callable, t_{nc} being the end of the non-call period; the maturity is always callable. The first callable date where $P_k \geq B_k^{\text{call}}$ is the call date τ.

Redemption at maturity. If the note is never called, it is redeemed at T at

$$R_T = N\left[(1 - \mathbf{1}_{\text{KI}}) + \mathbf{1}_{\text{KI}} \max\Big(\phi,\ \min\left(1,\ P_T/K_{\text{put}}\right)\Big)\right], \tag{1}$$

where $\mathbf{1}_{\text{KI}}$ is the knock-in event of the protection, K_{put} the strike of the short put and ϕ the capital floor. The knock-in event is

- *European protection*: $P_T < B^{\text{prot}}$;
- *American protection*: the index touches $B^{\text{prot}} S_{\text{ref}}$ at any time before maturity, continuously or on daily closes.

The standard note has $K_{\text{put}} = 1$ and $\phi = 0$: below the barrier, the investor receives NP_T and the redemption jumps from N to NB^{prot} at the barrier. The *airbag* note has $K_{\text{put}} = B^{\text{prot}}$: the redemption is continuous at the barrier and equal to NP_T/B^{prot} below it. A *capital guarantee* sets $\phi > 0$. Figure 1 draws the three profiles.

2.2 Phoenix

The Phoenix pays at each observation date $t_k \leq \tau$ (and $t_k \leq T$) the coupon

$$N\, c\, (t_k - t_{k-1})\, \mathbf{1}_{\{P_k \geq B_k^{\text{cpn}}\}} \tag{2}$$

and, with memory, the coupons of all the previous dates since the last payment. When the note is called, it is redeemed at N. With $B_k^{\text{cpn}} = 0$, the coupon is unconditional; with $B_k^{\text{cpn}} = B_k^{\text{call}}$ and memory, a coupon is paid only at the call date and its amount is $c\, t_k$: the Phoenix is then exactly a snowball Athena (Table 3 checks it numerically).

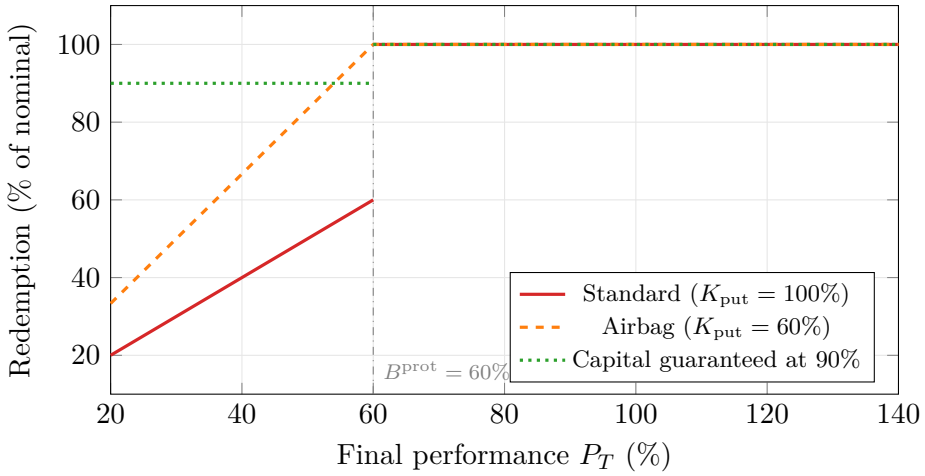


Figure 1: Redemption at maturity of a note that was never called, European protection at 60% (the last coupon is paid on top of it if $P_T \geq B_n^{\text{cpn}}$).

2.3 Parametrisation

Table 1 lists the parameters of the class `Phoenix`; every variant of Section 5 is one of these parameters changed.

Feature	**Parameter**	**Values**
Maturity, initial fixing, nominal	`maturity`, `initial_fixing`, `nominal`	years, index level, 100
Observation dates	`frequency` or `observation_dates`	monthly to annual, or any list ending at T
Coupon rate	`coupon_rate`	per annum, fraction of the nominal
Coupon condition	`coupon_barrier`, `memory`	fraction of S_{ref} (scalar or one per date), True / False
Autocall barrier	`autocall_barrier`, `step_down`	fraction of S_{ref}, decrease per observation
Non-call period	`first_call_date`	years
Capital protection	`protection_barrier`, `protection_type`	fraction of S_{ref}; `european`, `american`, `american_discrete`
Airbag, guarantee	`put_strike`, `floor`	fraction of S_{ref}, fraction of the nominal

Table 1: Parameters of the Phoenix (`Products/Phoenix/phoenix.py`).

3 Market Data and Model

The data are the CAC40 option chain of 12 February 2025 (spot $S_0 = 8042.19$, 142 quotes on 13 expiries from 9 days to 4.86 years) and the EURIBOR 6M zero-coupon curve. The forward curve $F(0,t)$ is bootstrapped from put–call parity with a piecewise constant borrow-plus-dividend yield $q(t)$ [2]; its integral reaches 0.136 at 4.86 years, so dividends lower the 4-year forward by about 11% with respect to a zero carry. Under $\mathbb{Q}$,

$$\frac{\mathrm{d}S_t}{S_t} = \partial_t \ln F(0,t)\,\mathrm{d}t + \sqrt{v_t}\,\mathrm{d}W_t^S, \qquad \mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \xi\sqrt{v_t}\,\mathrm{d}W_t^v, \qquad \mathrm{d}\langle W^S, W^v\rangle_t = \rho\,\mathrm{d}t, \tag{3}$$

with the parameters of Table 2. The notes are struck at the money: $S_{\mathrm{ref}} = S_0 = 8042.19$, and S_{ref} stays fixed when the Greeks bump the market. All the dates of the 4-year notes lie before the last listed expiry (4.86 years).

v_0	θ	κ	ξ	ρ	IV RMSE
0.0197	0.0438	1.308	0.482	-0.605	0.421 vol pts

Table 2: Heston parameters calibrated on the CAC40 surface of 12 February 2025 [2].

4 Pricing Method

4.1 Simulation

The paths of (S, v) are simulated with an Euler full truncation scheme on a weekly grid (52 steps a year, 208 steps for 4 years), the drift over each step being $\ln(F(0,t_{i+1})/F(0,t_i))$, so that $\mathbb{E}[S_t] = F(0,t)$ at every date. $N = 50{,}000$ paths with antithetic variates are used (seed 42); prices and standard errors are computed on the 25,000 independent pair averages. The observation dates are rounded to the grid; with weekly steps, every monthly, quarterly, semi-annual and annual date falls within half a week of its contractual date.

4.2 Cash flows and protection

For each path, the note is followed through its observation dates: at each date, the coupon is paid if the performance is above the coupon barrier (with the unpaid coupons if the note has a memory), then the note is called if the performance is above the autocall barrier. Each cash flow is discounted with $D(0,t_k) = e^{-r(t_k)t_k}$. The knock-in indicator of (1) is

- *European*: $\mathbf{1}_{\{P_T < B^{\text{prot}}\}}$;
- *American continuous*: replaced by its conditional probability given the simulated path, $1 - \prod_i (1 - p_i)$, where $p_i = \exp\big(-2\ln(S_{t_i}/H)\ln(S_{t_{i+1}}/H)/(v_{t_i}\Delta t)\big)$ is the probability that the Brownian bridge of $\ln S$ touches $H = B^{\text{prot}} S_{\text{ref}}$ between two grid dates [8];
- *American daily*: the barrier is checked on a daily grid (252 steps a year).

4.3 Variance reduction

Besides antithetic variates, the discounted European put struck at $B^{\text{prot}} S_{\text{ref}}$, whose Heston price is known semi-analytically (Lewis formula), is used as a control variate with an optimal coefficient estimated on the paths.

4.4 Fair coupon and scenario statistics

On a fixed set of paths, the call dates do not depend on the coupon, and every cash flow is affine in the coupon rate c: the price is $V(c) = V(0) + c\,(V(1) - V(0))$, and the coupon that prices the note at par is

$$c^\star = \frac{N - V(0)}{V(1) - V(0)}, \tag{4}$$

obtained with two valuations. It excludes the issuer margin and funding spread. The same paths give the probability of a call at each date, the probability to reach maturity, the probability of a capital loss ($\mathbf{1}_{\text{KI}} = 1$ and $P_T < K_{\text{put}}$), the expected loss at maturity, the expected life $\mathbb{E}[\min(\tau, T)]$ and the expected number of coupons.

4.5 Greeks

The Greeks are central finite differences with common random numbers [7], on a nominal of 100: Delta and Gamma (spot $\pm 1\%$, carry kept), Vega and Volga ($\sqrt{v_0}$ and $\sqrt{\theta}$ shifted by ± 1 vol point), Vanna, Rho (± 10bp on the zero curve, per $+1\%$), Dividend (± 10bp on $q(t)$, per $+1\%$) and Theta (dates moved by one day). Delta is given per index point: multiply by $S_0/100 = 80.4$ for the value change of a 1% move. Each Greek is reported with its Monte Carlo standard error.

4.6 Validation

Two limit cases have a closed form and one an exact counterpart (Table 3). In a flat Black–Scholes market ($S_0 = S_{\text{ref}} = 100$, $r = 3\%$, $q = 1\%$, $\sigma = 20\%$, $T = 2$, $N = 100$, Heston with $v_0 = \theta = \sigma^2$ and $\xi = 10^{-4}$, 10^5 paths):

- a Phoenix never called (autocall barrier out of reach) with unconditional quarterly coupons and a European protection at $B = 60$ is worth the sum of its discounted coupons plus $ND(T) - \frac{N}{S_{\text{ref}}}\left[\text{Put}(B) + (S_{\text{ref}} - B)\,\text{DigitalPut}(B)\right]$;
- a Phoenix called for sure at t_1 is worth $N(1 + c\,t_1)D(t_1)$.

On the CAC40, the Phoenix with memory and $B^{\text{cpn}} = B^{\text{call}}$ must have exactly the price of the snowball Athena with the same dates.

Case	**Reference**	**Monte Carlo**	**Std error**	**Error / std error**
Phoenix never called, unconditional 6% coupon (flat)	104.2404	104.2515	0.0141	0.79
Phoenix called for sure at $t_1 = 0.25$, $c = 6\%$ (flat)	100.742	100.742	0	exact
Phoenix with memory, $B^{\text{cpn}} = B^{\text{call}}$ vs snowball Athena (CAC40)	98.7669	98.7669	0.0585	identical paths

Table 3: Validation of the pricer.

5 Phoenix : Base case

Let us consider, under the risk-neutral measure $\mathbb{Q}$, the general case of a Phoenix in the base case, on a nominal of 1, with N observation dates, a coupon c per period, a coupon barrier B_c, an autocall barrier B_{ac} and a protection barrier B_p, with $B_p \leq B_c \leq B_{ac}$. As for the Athena, we denote by

$$E_i = \bigcap_{j=1}^{i-1} \{X_j < B_{ac}\}, \qquad i = 1, \dots, N, \qquad E_1 = \Omega, \tag{5}$$

the event that the product has not been called before date t_i. The event $\{\tau = t_i\}$ of an early redemption at date $t_i < T$ is then $E_i \cap \{X_i \geq B_{ac}\}$, and $\{\tau = T\} = E_N$. With memory, the coupon paid at t_i covers all the periods since the last payment: it is $c\, m_i$, with

$$m_i = i - \max(\{0\} \cup \{j < i : X_j \geq B_c\}), \tag{6}$$

and $m_i = 1$ without memory.

5.1 Price of the Phoenix

For a coupon c, the price of the Phoenix at inception is the sum of its discounted coupons and of its discounted redemption,

$$P_0 = \mathbb{E}^{\mathbb{Q}}\Big[\sum_{t_i \leq \tau} D(t_i)\, c\, m_i\, \mathbf{1}_{\{X_i \geq B_c\}} + D(\tau)\, \Pi_\tau\Big], \tag{7}$$

which, writing the autocall events explicitly, expands as

$$\begin{aligned} P_0 = &\sum_{i=1}^{N} D(t_i)\, c\, \mathbb{E}^{\mathbb{Q}}\big[m_i\, \mathbf{1}_{E_i \cap \{X_i \geq B_c\}}\big] \\ &+ \sum_{i=1}^{N-1} D(t_i)\, \mathbb{Q}(E_i \cap \{X_i \geq B_{ac}\}) \\ &+ D(T)\, \mathbb{Q}(E_N \cap \{X_N \geq B_p\}) \\ &+ D(T)\, \mathbb{E}^{\mathbb{Q}}\big[X_N\, \mathbf{1}_{E_N \cap \{X_N < B_p\}}\big]. \end{aligned} \tag{8}$$

The first term is the strip of conditional coupons, the second corresponds to an early redemption at date t_i, the third to the repayment of the notional at maturity, and the last one to the capital loss scenario. Contrary to the Athena, the coupon is not attached to the redemption: it is paid as long as the note is alive and the index above B_c, including in scenarios that end with a capital loss.

5.2 Linearity in the coupon and embedded put

The price of the Phoenix is also affine in the coupon: the coupon affects the amounts paid, never the redemption time τ nor the dates where a coupon is paid, which depend on the path of the underlying alone. On the event E_N, the terminal payoff can be rewritten, since $B_p < 1$, as

$$\Pi_T = 1 + c\, m_N\, \mathbf{1}_{\{X_N \geq B_c\}} - (1 - X_N)^+\, \mathbf{1}_{\{X_N < B_p\}}, \tag{9}$$

where the last term is again the down-and-in put with strike 100% and barrier B_p observed at maturity, sold by the investor. Substituting (9) into (8) yields the decomposition

$$P_0 = \underbrace{\mathbb{E}^{\mathbb{Q}}\big[D(\tau)\big]}_{\text{notional}} + \underbrace{c \sum_{i=1}^{N} D(t_i)\, \mathbb{E}^{\mathbb{Q}}\big[m_i\, \mathbf{1}_{E_i \cap \{X_i \geq B_c\}}\big]}_{B\ \text{(coupons)}} - \underbrace{D(T)\, \mathbb{E}^{\mathbb{Q}}\Big[(1 - X_N)^+\, \mathbf{1}_{\{X_N < B_p\}}\, \mathbf{1}_{E_N}\Big]}_{\text{PDI}^{ac}}, \tag{10}$$

that is

$$P_0 = A + c\,B, \qquad A = \mathbb{E}^{\mathbb{Q}}\big[D(\tau)\big] - \mathrm{PDI}^{ac}. \tag{11}$$

The zero-coupon part A does not depend on the coupon condition: for the same dates and the same autocall and protection barriers, *the Phoenix and the Athena have the same A*. They only differ through the value B of a unit coupon. Two limit cases bound it:

- with $B_c = 0$ (unconditional coupon), $m_i = 1$ and $B = \sum_i D(t_i)\,\mathbb{Q}(E_i)$, a strip of coupons knocked out at the call;
- with $B_c = B_{ac}$ and memory, $X_j < B_c$ for every $j < i$ on E_i, so that $m_i = i$ and B is exactly the B of the snowball Athena.

5.3 Fair coupon

As for the Athena, the fair coupon for a target price $P^\star$ (1, or $1 - m$ with an issuer margin m) is

$$\boxed{c^\star = \frac{P^\star - A}{B} = \frac{P^\star - \mathbb{E}^{\mathbb{Q}}\big[D(\tau)\big] + \mathrm{PDI}^{ac}}{B}} \tag{12}$$

and a single Monte Carlo simulation provides A and B. Since A is shared with the Athena of the same dates, the ratio of the two fair coupons is the inverse ratio of the coupon values,

$$\frac{c^\star_{\text{Phoenix}}}{c^\star_{\text{Athena}}} = \frac{B_{\text{Athena}}}{B_{\text{Phoenix}}}. \tag{13}$$

On the base case, the quarterly snowball Athena pays 10.54% and the Phoenix 6.04%: a unit coupon is worth 1.75 times more in the Phoenix, because it is also paid in the scenarios where the index stays between B_c and B_{ac}.

5.4 Remarks on the embedded put and on the coupons

The put PDI^{ac} is the same as in the Athena: knocked out by the autocall, bounded by the European down-and-in put, which decomposes into a vanilla put struck at the barrier and a digital put,

$$(1 - X_N)^+\,\mathbf{1}_{\{X_N < B_p\}} = (B_p - X_N)^+ + (1 - B_p)\,\mathbf{1}_{\{X_N < B_p\}}. \tag{14}$$

The coupons add a second exposure to the shape of the distribution: each coupon is a digital call struck at B_c, knocked out by an earlier call. Without memory, $B = \sum_i D(t_i)\,\mathbb{Q}(E_i \cap \{X_i \geq B_c\})$; the memory can only increase it, up to $\sum_i i\,D(t_i)\,\mathbb{Q}(E_i \cap \{X_i \geq B_c\})$. Both strips are sensitive to the skew around B_c.

5.5 Structuration :

From the investor's perspective, the basic Phoenix note is composed of:

- a long position in an *autocallable zero-coupon note* with maximal maturity T, which repays the notional at the first observation date t_i such that $X_i \geq B_{ac}$, and at maturity otherwise;
- a long position in a *strip of digital coupons* paying c (or $c\,m_i$ with memory) at each date t_i where $X_i \geq B_c$, knocked out when the product is autocalled;
- a short position in a *down-and-in put* (PDI) with strike 100% and barrier B_p observed at maturity, which is knocked out whenever the product is autocalled.

5.6 Sensitivity analysis :

We will now look at the different sensitivities of the product and what are the impacts of a move on certain parameters on c. We are now thinking from the investor point of view and with an already contracted note.

Volatility

The investor is short the PDI, whose price increases with volatility: he is short vega. The digital coupons, struck at $B_c = 70\%$ below the spot, also lose value when the volatility rises, since the index is less likely to stay above the coupon barrier. The base note loses 0.33 per vol point.

If the product is not contracted yet, a higher volatility makes the put more expensive, and the fair coupon c increases.

Protection barrier

If the protection barrier is higher, then it is more probable to cross it and it reduces the price of the note. The effect on the coupon c is that it is increased: 5.16% at 50%, 6.04% at 60%, 6.85% at 70%.

Recall Barrier

If the recall barrier is increased, it is less probable to be recalled before maturity: the investor keeps the put longer, and the fair coupon c increases (5.46% with an autocall barrier at 90%, 6.04% at 100%). Contrary to the Athena, a longer life also means more coupons, which limits the effect; a step-down of the autocall barrier, which shortens the coupon stream, raises the fair coupon (6.20%).

Coupon barrier

If the coupon barrier is increased, fewer coupons are paid and the fair coupon c increases: 5.43% at 50%, 6.04% at 70%, 6.67% at 80%, and 5.13% for an unconditional coupon.

Memory

The memory pays back the missed coupons as soon as the index comes back above B_c: it is worth 0.37 point of coupon at a coupon barrier of 70% (6.04% instead of 6.41% without memory), and 2.41 points at 90%.

Dividend

If the underlying asset increases its dividend, the forward decreases: the put is more valuable, the coupons and the calls less likely, and the fair coupon c increases. The investor does not receive the dividends. The exposure is a third of the Athena one (-0.32 per point of carry against -1.12), the coupons being paid on a shorter horizon.

Spread

Risk free rate

Maturity

Skew

Correlation

5.7 Pricing : A numerical example

Let us now consider a synthetic Phoenix issued by a risky issuer, which requires taking into account the credit spread of the issuer. The product is written on the CAC 40 index (dividends not reinvested). Its initial level is fixed on 31 January 2025 and it matures in November 2027. The underlying is observed quarterly: at each observation date, a coupon c is paid if the index closes at or above the coupon barrier $B_c = 70\%$ of its initial level, together with the coupons missed since the last payment (memory), and the product is redeemed early if the index closes at or above 100% of its initial level. If the product has not been called, the investor receives at maturity the last coupon if the index is at or above the coupon barrier, the notional if it is at or above the protection barrier $B_p = 60\%$, and the notional reduced by the full decline of the index otherwise.

5.7.1 Model

5.7.2 Pricing

5.7.3 Sensitivities

5.8 Features of Phoenix products

5.8.1 Exercise frequency

5.8.2 Memory

5.8.3 Step-down barrier

5.8.4 Airbag

5.8.5 Buffer

5.8.6 Guaranteed capital

5.8.7 Coupon range accrual

5.8.8 Duo mechanism

5.8.9 Look back

The base Phoenix has a maturity of 4 years, quarterly observations, a coupon of 6% a year (1.5% a quarter) with memory and a coupon barrier at 70%, an autocall barrier at 100% and a European protection at 60%. Its price is 99.963 ± 0.099 (95%) and its fair coupon 6.04% a year. 55.4% of the notes are called after one quarter (Figure 2), the expected life is 1.03 years, 3.28 coupons are paid on average, 13.3% of the notes reach maturity and 5.2% lose capital, with an expected loss of 3.05%.

5.9 All the forms of the Phoenix

Table 4 changes one feature of the base case at a time. All prices are computed at the same coupon of 6%; the fair coupon is the comparable quantity.

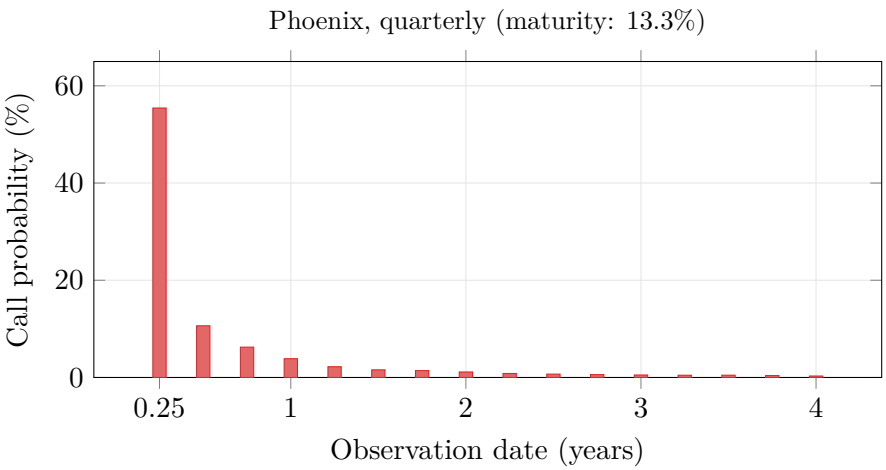


Figure 2: Base Phoenix: probability of being called at each observation date (autocall barrier at 100%).

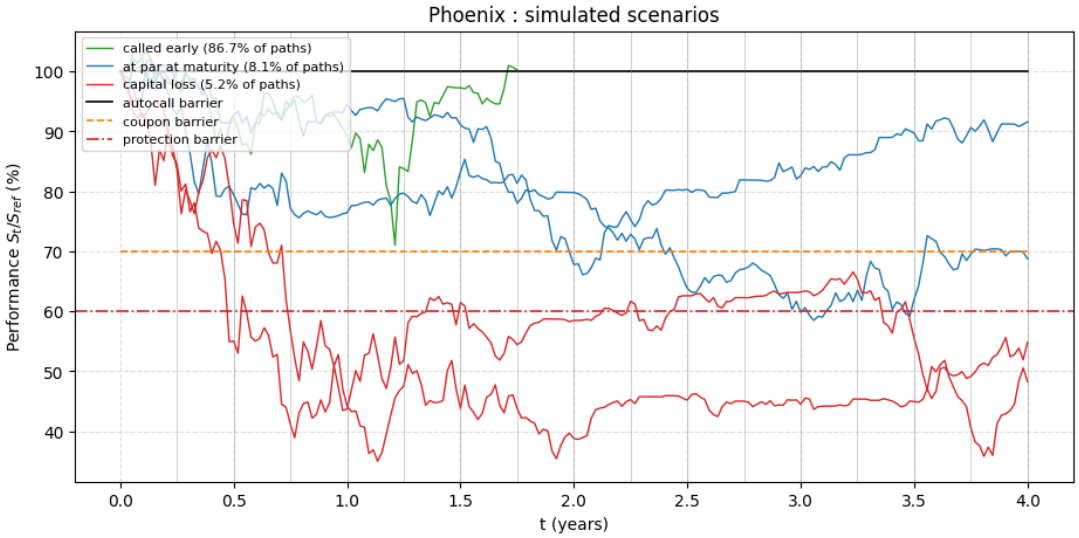


Figure 3: Base Phoenix: simulated paths of the performance, grouped by outcome.

Reading the table.

- *Coupon barrier and memory.* Lowering the coupon barrier from 80% to 50% raises the expected number of coupons from 2.80 to 3.83 and lowers the fair coupon from 6.67% to 5.43%; an unconditional coupon gives 5.13%. Memory is worth 0.37 point of coupon at a barrier of 70% (6.04% instead of 6.41%), and its value grows with the barrier (Figure 4): at 90%, 7.87% with memory against 10.28% without. With the coupon barrier equal to the autocall barrier, the memory Phoenix is the snowball Athena (10.54%, as the quarterly Athena of the companion report [1]).

- *Observation frequency and early calls.* Contrary to the Athena, a Phoenix pays its coupons as long as it lives: everything that shortens its life shortens the coupon stream. Monthly observations raise the fair coupon (6.19%) and annual observations lower it (5.63%); a step-down of the autocall barrier raises it (6.20%), whereas it lowers the Athena coupon. A non-call period of one year lengthens the expected life to 1.89 years and lowers the fair coupon to 5.46%.

- *Protection.* The American protection costs 1.14 points of coupon (7.18%), the airbag gives

Variant	Price at 6%	Fair coupon (% p.a.)	P(maturity) (%)	P(loss) (%)	E[loss] (%)	E[life] (y)	E[coupons]
Base (quarterly, memory, CB 70%)	99.96	6.04	13.29	5.22	3.05	1.03	3.28
No memory	99.67	6.41	13.29	5.22	3.05	1.03	3.28
Coupon barrier 50%	100.54	5.43	13.29	5.22	3.05	1.03	3.83
Coupon barrier 60%	100.30	5.67	13.29	5.22	3.05	1.03	3.61
Coupon barrier 80%	99.49	6.67	13.29	5.22	3.05	1.03	2.80
Coupon barrier = autocall barrier (= snowball Athena)	97.80	10.54	13.29	5.22	3.05	1.03	0.87
Unconditional coupon (CB 0%)	100.87	5.13	13.29	5.22	3.05	1.03	4.13
Monthly observations	99.91	6.19	7.77	3.13	1.83	0.59	5.53
Semi-annual observations	100.22	5.82	18.61	6.97	4.08	1.48	2.38
Annual observations	100.62	5.63	25.70	9.06	5.27	2.06	1.67
Autocall barrier 90%	100.19	5.46	3.13	1.70	1.01	0.42	1.37
Step-down 1.25% per quarter	99.86	6.20	7.14	4.29	2.54	0.84	2.62
Non-call 1 year	100.86	5.46	22.06	8.42	4.92	1.89	6.21
Protection 50%	100.71	5.16	13.29	3.43	2.25	1.03	3.28
Protection 70% (= coupon barrier)	99.28	6.85	13.29	7.32	3.78	1.03	3.28
American protection (continuous)	99.00	7.18	13.29	9.22	4.11	1.03	3.28
American protection (daily)	99.07	7.11	13.23	8.98	4.06	1.03	3.28
Airbag	101.28	4.49	13.29	5.22	1.61	1.03	3.28
Capital guaranteed 90%	102.26	3.33	13.29	5.22	0.52	1.03	3.28

Table 4: Phoenix variants on the CAC40 (Heston, 50,000 paths, standard errors of the prices between 0.02 and 0.06). P(call t_1) is 55.4% for every quarterly variant except the non-call period (0%) and the 90% autocall barrier (90.0%).

back 1.55 points (4.49%), a 90% guarantee leaves 3.33%.

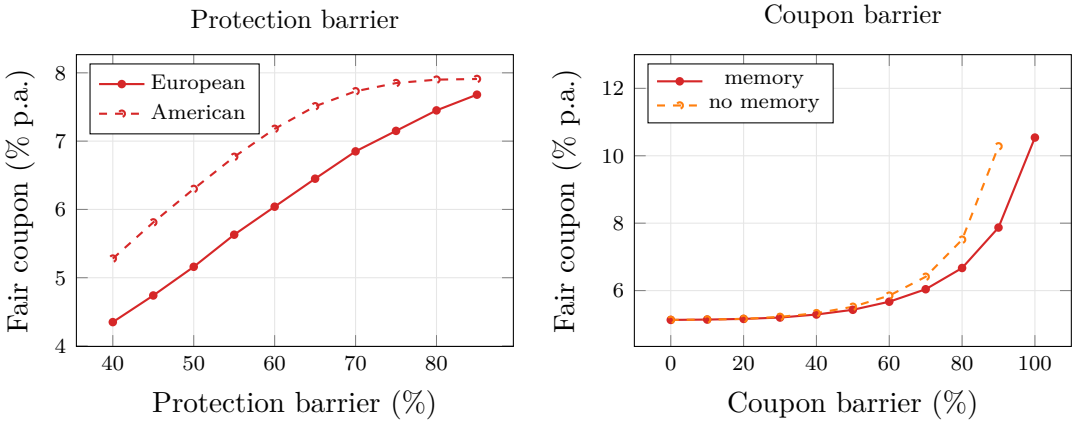


Figure 4: Phoenix: fair coupon as a function of the barriers (other features of the base case unchanged; American: continuous monitoring). Without memory, the fair coupon reaches 23.94% at a coupon barrier of 100% (off scale).

5.10 Greeks

Table 5 gives the Greeks of five Phoenix variants.

The Phoenix is almost Delta-neutral at the money (0.047 for a 1% move against 0.41 for the Athena): a rise of the index makes the coupons more likely but also calls the note earlier, which stops a coupon stream priced above par. When the coupon is well above its fair level, the second effect dominates and the Delta becomes negative (airbag and unconditional coupon, whose prices are 101.3 and 100.9). The Phoenix is short volatility (-0.33 per vol point) and short dividends (-0.32 per point, a third of the Athena): its coupons are paid on a shorter horizon, so the forward matters less. Figures 5 and 6 display the Greeks.

Sensitivities to the Heston parameters. The base Phoenix has $\partial V/\partial v_0 = -40.4$, $\partial V/\partial\theta = -44.9$, $\partial V/\partial\kappa = -0.19$, $\partial V/\partial\xi = 1.36$ and $\partial V/\partial\rho = 0.87$: 58% to 65% of the Athena exposures

Phoenix	**Price**	**Delta** (per point)	**Delta 1%** (per 1% move)	**Vega** (per vol pt)	**Vanna** ($\times 10^4$)	**Rho** (per 1%)	**Dividend** (per 1%)	**Theta** (per day)
Base	99.96	0.00059	0.047	−0.335	6.13	−0.599	−0.325	0.0050
No memory	99.67	0.00106	0.085	−0.365	6.27	−0.560	−0.357	0.0053
American protection	99.00	0.00229	0.184	−0.394	5.72	−0.448	−0.437	0.0039
Airbag	101.28	−0.00151	−0.121	−0.224	5.38	−0.875	−0.101	0.0026
Unconditional coupon	100.87	−0.00077	−0.062	−0.253	5.67	−0.745	−0.202	0.0048

Table 5: Greeks of Phoenix variants (nominal 100, coupon 6%). Base case standard errors: Delta 0.00017, Vega 0.012, Vanna 0.8×10^{-4}, Rho 0.044, Dividend 0.044, Theta 0.0014; Gamma and Volga are not significant.

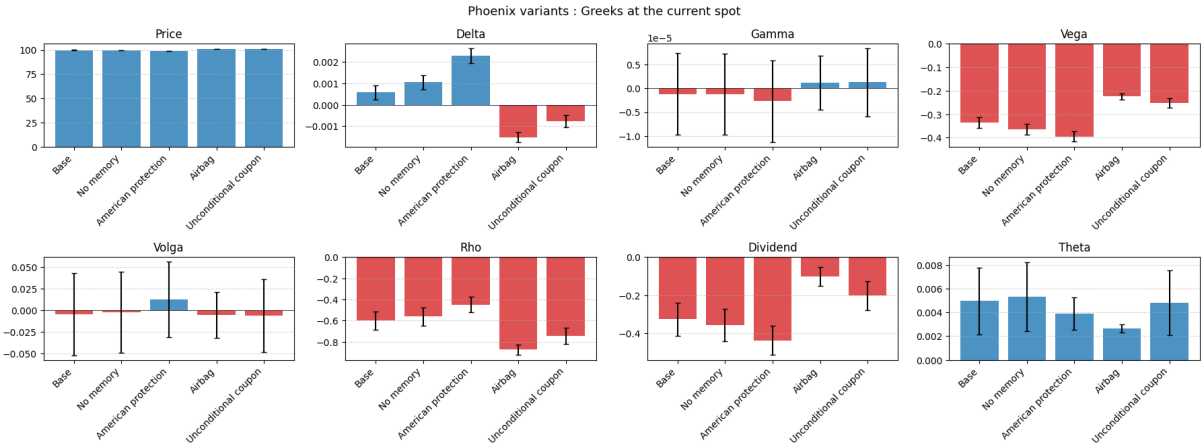


Figure 5: Greeks of the Phoenix variants with 95% confidence intervals.

to the variance level.

6 Phoenix versus Athena

	Athena (annual, snowball)	**Phoenix** (quarterly, memory, CB 70%)
Fair coupon (% p.a.)	9.41	6.04
Probability of a call at the first date (%)	55.69	55.43
Expected life (years)	2.06	1.03
Probability of a capital loss (%)	9.06	5.22
Expected loss (% of nominal)	5.27	3.05
Delta (per 1% move)	0.407	0.047
Vega (per vol point)	−0.549	−0.335
Dividend (per 1% of carry)	−1.115	−0.325

Table 6: Base cases compared (European protection at 60%, autocall at 100%, 4 years); Athena figures from the companion report [1].

The Athena concentrates its coupons in the scenarios where the index is up, which is why its fair coupon is higher and its Delta almost nine times larger; the Phoenix pays regularly as long as the index stays above 70%, a more stable income for a lower rate. With the same quarterly dates, the snowball Athena and the memory Phoenix with $B^{\text{cpn}} = B^{\text{call}}$ have the same fair coupon (10.54%), which is the identity of Section 2.

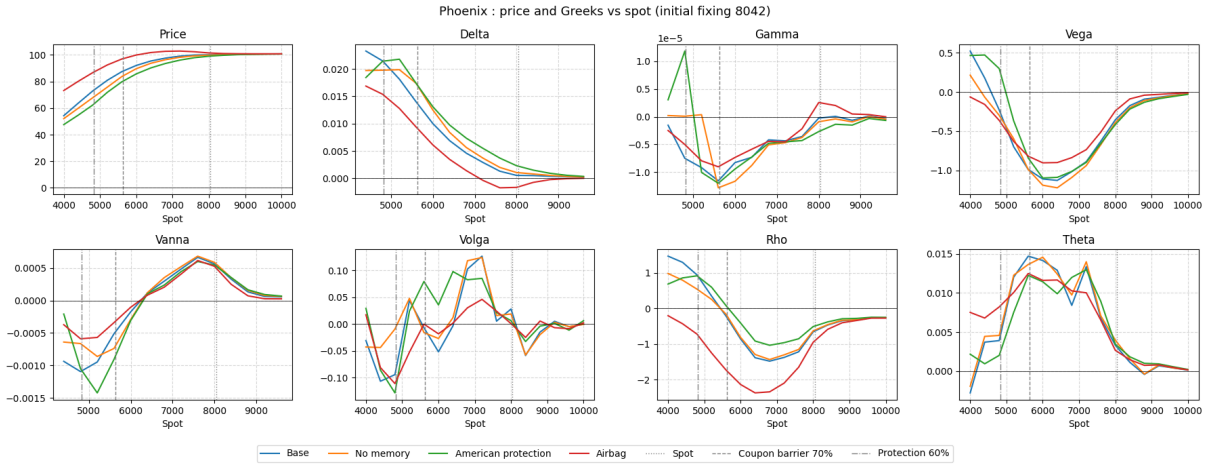


Figure 6: Phoenix: price and Greeks as functions of the spot (30,000 paths, spot step 400).

7 Implementation

7.1 Code

- `Products/Phoenix/phoenix.py`: the self-contained class `Phoenix` (contract, `path_payoff`, `fair_coupon`, `statistics`);
- `Equity_Models/monte_carlo.py`: `MonteCarloPricer`, which simulates the paths and calls `path_payoff` for products with several cash flows;
- `Equity_Models/Heston.py`, `Equity_Models/forward_curve.py`: calibrated model and forward curve;
- `Equity_Models/greeks.py`: Greeks with common random numbers, including the Dividend Greek;
- `Products/market.py`: CAC40 market data and calibrated Heston model (`cac40_heston`);
- `Products/notes_analysis.py`: comparison of variants and plots;
- notebook `Products/Phoenix/phoenix_pricer.ipynb`.

```
from market import cac40_heston
from phoenix import Phoenix
from Equity_Models.monte_carlo import MonteCarloPricer
from Equity_Models.greeks import Greeks

mkt = cac40_heston()
pricer = MonteCarloPricer(n_paths=50_000, steps_per_year=52, seed=42)
note = Phoenix(4.0, mkt.S0, 0.06, coupon_barrier=0.7, memory=True, frequency
    =0.25,
               autocall_barrier=1.0, protection_barrier=0.6, protection_type="
                   european")
print(pricer.price(mkt.heston, note))              # price and standard error
print(note.fair_coupon(pricer, mkt.heston))        # coupon p.a. at par
print(note.statistics(pricer, mkt.heston))         # call probabilities, loss,
    life, coupons
print(Greeks(pricer).compute(mkt.heston, note, Greeks.ORDER))
```

7.2 Cost

With N paths and M steps, the simulation costs $\mathcal{O}(NM)$ in time and stores S and v, i.e. $16(M+1)N$ bytes: 167 MB for $N = 50{,}000$ and the weekly grid of 4 years, 807 MB on a daily grid. Following the note through its n observation dates costs $\mathcal{O}(nN)$, the American continuous protection $\mathcal{O}(NM)$. Measured times (one CPU core): 0.30 s for a valuation including the simulation; 0.02 to 0.12 s for the price, the fair coupon and the statistics of a variant on cached paths (1.5 s with an American protection); 26 s for the eight Greeks of five notes (14 valuations each); 41 to 44 s for the spot profiles of four notes (16 spots $\times$ 6 scenarios, 30,000 paths).

8 Limitations and Next Steps

- *Single underlying.* Most autocalls sold today are worst-of on two to four indices or stocks; they need a multi-asset model and a correlation calibration.
- *Model.* The value of an autocall depends on the forward skew and on the joint dynamics of spot and volatility between observation dates; a pure Heston model with constant parameters is not the market standard, which is a local stochastic volatility model calibrated to the whole surface [6].
- *Fair coupon.* It prices the note at par without issuer margin, funding spread or issuer credit risk; a real issue price is below par by the margin.
- *Discretisation.* Euler full truncation with the Feller condition violated ($2\kappa\theta/\xi^2 = 0.49$), weekly grid with the variance frozen over each week for the Brownian bridge.
- *Greeks.* Gamma and Volga are not significant with 50,000 paths: the payoff jumps at the coupon, autocall and protection barriers.
- *Dividends.* Continuous dividend yield bootstrapped from the forwards; real notes are exposed to discrete dividends and to dividend risk, which the Dividend Greek only measures for a parallel shift.

9 Conclusion

A single parametrisation covers the Phoenix with all its usual features, and a Monte Carlo pricer under a calibrated Heston model recovers two exact limit cases within one standard error and the exact identity with the snowball Athena. On the CAC40 of 12 February 2025, the base 4-year Phoenix pays 6.04% a year; its zero-coupon part is the same as the Athena one, and its lower coupon only reflects that a unit coupon is worth 1.75 times more. The coupon barrier and the protection are the main drivers of the fair coupon, the memory is worth 0.37 point at a barrier of 70%, and everything that shortens the life of the note (more frequent observations, step-down) raises the fair coupon. The note is almost Delta-neutral at the money, short volatility and short dividends, a third of the Athena on dividends.

A Notation

Symbol	Meaning
N, S_{ref}, T	Nominal, initial fixing, maturity
t_k, $\delta_k = t_k - t_{k-1}$	Observation dates and accrual periods
$P_k = S_{t_k}/S_{\text{ref}}$	Performance at t_k
B_k^{call}, B_k^{cpn}, B^{prot}	Autocall, coupon and protection barriers (fractions of S_{ref})
m_i	Number of coupon periods paid at t_i (memory)
c, $c^\star$	Coupon rate per annum, fair coupon
K_{put}, ϕ	Strike of the short put, capital floor
τ	Call date
$\mathbf{1}_{\text{KI}}$	Knock-in event of the protection

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